

Cold-start method for proton-exchange membrane fuel cells based on locally heating the cathode

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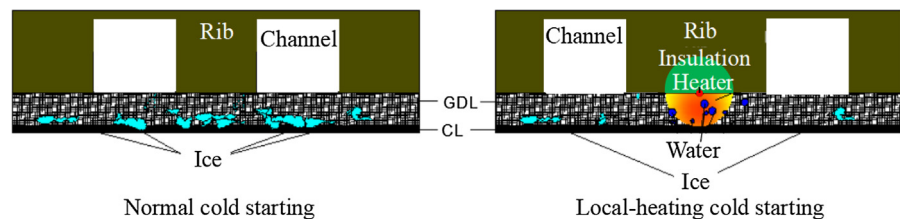
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HIGHLIGHTS

- The effect of local heating on cold-start performance is investigated.
- The wires contact the cathode GDL to directly heat the membrane electrode.
- Concentrated (one wire) and dispersed (three wires) local heating are compared.
- One heating wire more effectively improves cold-start performance than three wires.

GRAPHICAL ABSTRACT

Local heating apparently improves the cold-start ability of the fuel cell; the greater the local heating power, the faster the temperature of the reaction zone increases; the fuel cell using one heating wire showed better cold-start performance than that using three heating wires.



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ABSTRACT

The key to successfully cold starting proton-exchange membrane fuel cells is increasing the cell temperature above 0 °C before the electrochemical reaction stops, because ice forms in fuel cells below this temperature. To decrease the input of external heat energy to fuel cells during cold starting, this paper proposes a local-heating method to improve the cold-start performance of fuel cells. During the experiments, heating wires were placed under partial ridges in the cathode plate to improve the cold-start performance of the fuel cells. The cold-start characteristics of the locally heated fuel cells were analyzed by measuring the voltage, high-frequency impedance, and cathode (gas diffusion layer) temperature for different heating power densities and number of heating wires. The results show that locally heating the cathode improves the cold-start capability of the fuel cell, and increasing the heating power density to heat the fuel cell enhances the voltage stability during cold starting of the cell. Furthermore, at a constant heating power density, the fuel cell using one heating wire shows better cold-start performance than that heated using three heating wires.

1. Introduction

Proton-exchange membrane fuel cells (PEMFCs) are widely used for their cleanliness and high energy conversion rates [1]. Simultaneously, improving the cold-start ability of the PEMFCs used in vehicles is important to the application of such cells in environments below 0 °C. Ice formation in these PEMFCs during cold starting separates the catalytic

layer (CL) and gas diffusion layer (GDL) and decreases the reaction areas [2], thereby lowering or even stopping the electrochemical reaction rate [3]. After multiple freeze–thaw cycles in the PEMFCs, platinum particles in the CL are removed, and the fuel-cell impedance increases [4]. In addition, volume expansion owing to water freezing in the cathode membrane electrode causes other problems in the cells. Specifically, it breaks off the carbon fiber in the GDL [5], increases the

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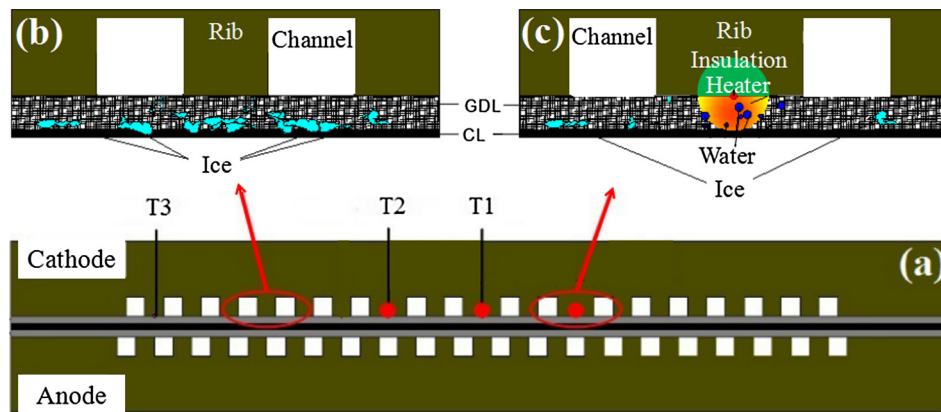


Fig. 1. Local heating principle underlying the cold starting of fuel cells: (a) location of heating wires and thermocouples; (b) cold starting without heating; (c) cold starting with local heating.

width and length of cracks in the microporous layer (MPL) [6], and damages the surface of the proton-exchange membrane (PEM) [7].

To successfully cold start fuel cells in a cold environment (below 0 °C), Florian et al. [8] used methanol as a cold-start antifreeze for PEMFCs; however, methanol impairs the normal temperature performance of fuel cells. Jiang et al. [9] and Ko et al. [10] studied the effect of current density on the cold-start performance. They believed that increasing the initial current density not only accelerates the increase in fuel-cell temperature but also increases water accumulation in the cathode CL, easily resulting in cold-start failure. Amamou et al. [11] proposed a real-time adaptive cold-start strategy of fuel cells to control the operating current in real time and quickly cold start fuel cells. Luo et al. [12] and Lin et al. [13] determined that the more the number of single PEMFC, the faster the temperature of the fuel-cell stack rises, and the better is the cold-start performance. Although the PEMFC can generate more heat energy when operating at a normal temperature [14], when the ambient temperature is low, utilizing the heat generated by an electrochemical reaction to increase the fuel-cell temperature for a successful cold start is difficult [15]. Phase-change materials can store a large amount of latent heat [16] and can be used as microscale heat-transfer materials [17]. Therefore, for short-time shutdown of fuel cells, the latent heat of phase change can be used as a heat source. Using numerical calculations, Sasmito et al. [18] found that fuel-cell stacks can be successfully started from −20 °C when a phase-change material is used as a heat-storage medium. Furthermore, external heating sources, which mainly include a circulating heating medium [18], heated reaction gas [19], thermoelectric device [20], and external combustion, can help effectively cold start PEMFCs [21]. Zhang et al. [22] also proposed a cold-start method in which PEMFCs are heated using a heating medium circulating around the fuel cells. Li et al. [23] wrapped heat insulation materials around a fuel-cell stack while circulating a heating medium to preheat the stack during cold starting. Although these methods heat the PEMFCs, the use of a circulating heating medium can result in a steady increase in the fuel-cell temperature; these methods also consume a large amount of heat energy and prolong the cold-start process because the circulating heating medium and fuel cells simultaneously consume heat energy. In addition, to decrease the volume of the fuel-cell heat-exchange system, Guo et al. [24] and Jiao et al. [25] showed that using air as a medium increases the intake gas temperature and decreases the current density loss in fuel-cell cold-start simulation. Nguyen et al. [26] proposed a heat-recovery system by preheating the intake air, obviously improving the heat-utilization rate for heating the fuel-cell stack. Khandelwal et al. [27] estimated that heating anode gas has a high heat-utilization rate as hydrogen has a larger heat transfer coefficient than air. However, heating the reaction gas to increase the fuel-cell temperature still utilizes a large amount of heat energy and prolongs the cold-start time as the reaction gas has a

low heat capacity. Although external combustion can quickly increase the temperature of PEMFCs [28], the use of a burner considerably increases the volume and weight of the fuel-cell system and results in uneven fuel-cell temperature [29]; such a system is unsuitable for cold starting PEMFCs in vehicles. Zhou et al. [30] used an electric heater to heat a fuel-cell stack during a simulated cold-start process and were able to successfully cold start the stack at −20 °C. Thompson et al. [31] proposed a patent to place heating wires in PEMFCs to help cold starting but did not conduct corresponding experimental investigations.

The existing heating modes for cold starting PEMFCs were comprehensively analyzed. Although the electrical-heating-based cold-start mode has many advantages, such as flexible control, simple add-on devices, and high heating efficiency, the direct use of heating wires for heating PEMFCs during the cold-start process has rarely been studied experimentally. In particular, the local-heating-based cold-start modes lack corresponding cold-start experimental and theoretical analyses. Therefore, in this study, a local-heating-based cold-start method, in which heating wires are placed under cathode ridges, was experimentally analyzed to successfully cold start PEMFCs while inputting less external heat energy. The local heating method is used to rapidly increase the temperature of the partial membrane electrode above 0 °C using a small amount of heat energy to maintain a continuous electrochemical reaction in this region. Then, the reaction heat released in this partial region can further increase the temperature in other reaction zones above 0 °C to successfully cold start fuel cells. Changes in the voltage, impedance, and cathode GDL temperature were analyzed to study the effect of the local heating method on the cold-start capability of PEMFCs for various heating power densities and different numbers of heating wires in the fuel cell.

2. Experimental devices and system

2.1. Design of the locally heated fuel cell

Fig. 1 shows the locations of heating wires placed under ridges in the cathode and the local heating principle underlying the cold starting of fuel cells. The central reaction zone of a PEMFC has the highest current density in the fuel cell that easily causes a vacuum effect from starvation [32] and is crucial to a successful cold start [33]. To heat the membrane electrode assembly (MEA) using external heat energy, we placed heating wires in the center of the fuel cell. In addition, to compare the effects of concentrated and dispersed local heating and to prevent the shape of the bipolar plate from affecting the fuel-cell performance, different arrangements of heating wires were employed. Specifically, either one wire was employed in the center of the fuel cell or three wires were employed, i.e., one in the center and two identical heating wires under the ridges on either side (Fig. 1(a)). Fig. 1(c) shows

mini channels under three wide ridges in the cathode bipolar plate; these channels are filled with a heat-insulation material. The heating wires ($\text{Cr}_{20}\text{Ni}_{80}$) placed on the surface of the heat insulation material contact the cathode GDL to directly heat the membrane electrode. As shown in Fig. 1(c), the membrane electrode heated by the heating wire forms a local high-temperature region relative to the unheated membrane electrode during the cold-start process. In addition, ice is not formed in the local high-temperature region, and the partial reaction region can continue to undergo the electrochemical reaction. In contrast, the temperature in the reaction zone without heating remains below 0°C for a long time, and ice continuously accumulates on the cathode CL surface, thereby decreasing the reaction area, as shown in Fig. 1(b). The widths of the three ridges with the heating wires were increased to 2 mm due to processing limitations, while the widths of the other ridges were 1 mm. The local heating may result in a temperature difference, which can be measured through thermocouples in the membrane electrode. These K-type thermocouples (diameter of 0.25 mm) were used to measure the surface temperature of the cathode GDL through small holes in the cathode bipolar plate, as shown in Fig. 1(a). This figure also shows the locations of the thermocouples T1, T2, and T3 with respect to the heat-source symmetry. T1 and T2 measured the temperature of the cathode GDL near the heating wires, while T3 measured the temperature at the edge of the cathode GDL. As the T3 temperature is the lowest in the overall reaction zone, the fuel cell can be considered to be successfully cold started when this temperature exceeds 0°C .

Fig. 2 shows the structure of the locally heated fuel cell used in this study. In the experiment, the reaction area of the fuel cell was 25 cm^2 , and the widths of the straight channels were 1 mm each. Both the cathode and anode GDLs (TGP-060) were $190\text{-}\mu\text{m}$ thick and the MPL was $50\text{-}\mu\text{m}$ thick. The PEM was made of type Nafion 212, and both the anode and cathode had the same platinum loading of 0.4 mg cm^{-2} in the CL. Every heating wire had a diameter of 0.15 mm and length of 47 mm, while the semicircular heat-insulation layer had a radius of 0.5 mm. The heating wires were clamped in the GDL and heat-insulation layer in the cathode, and the wires supplying the electrical power to the heating wires through the polar plate, current collector, and end plate were connected to the DC power supply. In addition, the anode and cathode comprised a heat-insulation layer between the current collectors and end plates, respectively, to decrease heat loss in the reaction zone.

2.2. Experimental system

Fig. 3 shows the experimental system for cold starting the locally heated fuel cell. The electrochemical workstation (Interface 5000E) whose frequency was 2500 Hz was used to measure the fuel-cell impedance during the cold-start process, and its lowest scanning time was

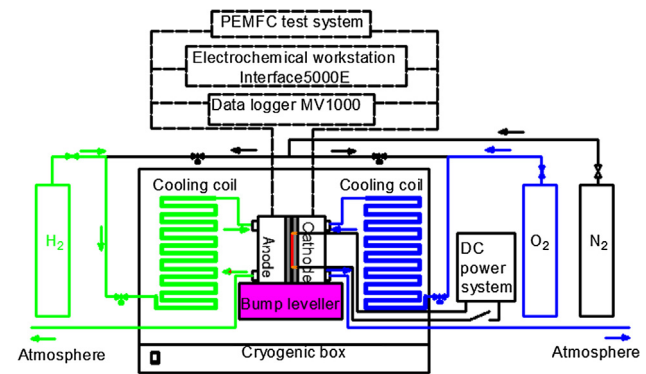


Fig. 3. Cold-start schematic of locally heated fuel cell.

$10\text{ }\mu\text{s}$. Before entering the fuel cell, both hydrogen and air were cooled to the experimental temperature (-10°C) by cooling the coils in the cryogenic box (Fig. 3). The DC power supply (ITECH-IT6720) could supply a maximum power of 100 W to the heating wires. In addition, the control accuracies of the output voltage ($0\text{--}60\text{ V}$) and output current ($0\text{--}5\text{ A}$) were $< 0.05\%$ and $< 0.2\%$, respectively. A data-acquisition recorder (YOKOGAWA-MV1000) was used to measure and record the cathode GDL temperature, and the measuring accuracy of the thermocouples (K-type) was $< 0.1^\circ\text{C}$. The fuel-cell test system (BATE-FC100) had a voltage measurement range of $0\text{--}2\text{ V}$, and its accuracy was $\pm 0.1\%$ FS. After fuel-cell operation, any water remaining in the fuel cell can seriously affect the next cold-start performance [34]. Therefore, the fuel cell was purged using nitrogen [35] for 1 h at 25°C before cold starting to remove water from the cathode and anode sides of the fuel cell [36]. After purging, the fuel cell was cooled to the experimental temperature (-10°C) and continued cooling for 3 h. At the beginning of the cold-start process, the heating wires in the cathode were powered simultaneously, and the voltage, impedance, and cathode GDL temperature were tested.

As the ridges with the heating wires were wider than the other ridges, the ability to remove water may be decreased. In addition, the heat-insulation material filled in the wide ridges could increase the contact resistance between the GDL and bipolar plate in the cathode. Therefore, analyzing the effect of the locally heated bipolar plate on the normal temperature performance of the fuel cell before cold starting is important. Table 1 lists some experimental conditions for analyzing the normal-temperature and cold-start performance of the locally heated fuel cell.

Table 1

Experimental conditions for fuel cells operated at normal temperature and cold starting.

Types of cathode polar plate	Normal/Local heating	Measurement units
Normal temperature performance		
Hydrogen flow	1800	sccm
Air flow	5000	sccm
Fuel-cell temperature	60	$^\circ\text{C}$
Relative humidity of hydrogen	100	%
Relative humidity of air	100	%
Cold-start performance		
Starting temperature	-10	$^\circ\text{C}$
Starting current density	0.04	A cm^{-2}
Number of heating wires	1/3	
Heating power density P1/P2/P3/ P1/P2	47.0/52.2 /60.6/ 365.0/438.0	W kg^{-1}
Relative humidity of hydrogen	0	%
Relative humidity of air	0	%

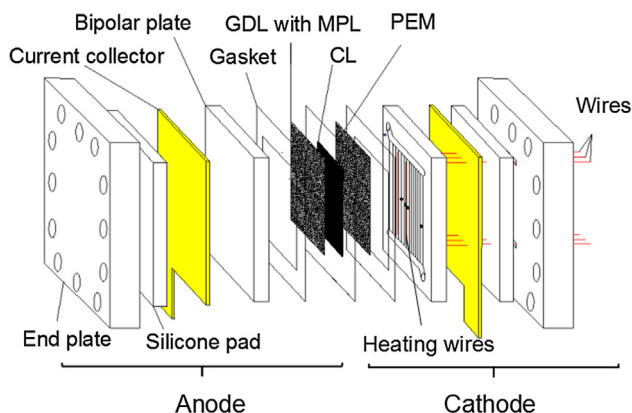


Fig. 2. Structure of the locally heated fuel cell.

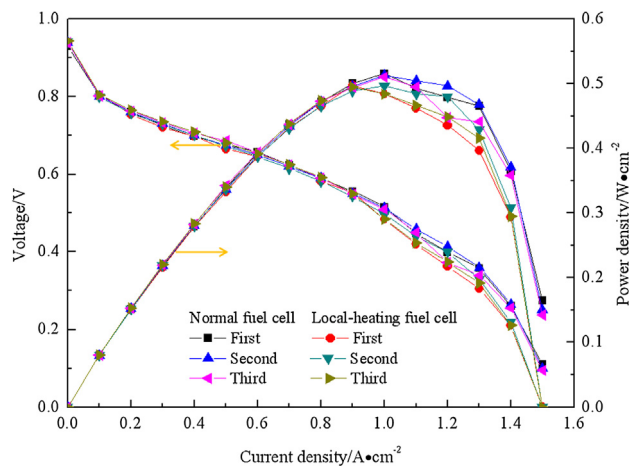


Fig. 4. Effect of cathode-type bipolar plate on the fuel-cell performance at 60 °C.

3. Experimental results and discussion

3.1. Effect of locally heated bipolar plate on the normal temperature performance of the PEMFC

As the structure of the cathode bipolar plate changed after installing heating wires, experiments were conducted to analyze the current-voltage (I-V) performance (60 °C) of the locally heated fuel cell with a locally heated bipolar plate before cold starting. The voltage and power density of the PEMFC using two bipolar plates (normal and locally heated bipolar plates) changes with increasing current density, as shown in Fig. 4. Two fuel cells were tested three times each with a maximum experimental error of no more than 3%. The type of the cathode bipolar plate had little effect on the fuel-cell output voltage until the current density reached 0.9 A cm^{-2} . After the current density exceeded 1.0 A cm^{-2} , the output voltage of the fuel cell using the bipolar plate with heating wires was lower than that of the fuel cell using the normal bipolar plate. As three wide ridges existed in the locally heated bipolar plate, the path through which air reached the cathode CL under the wide ridges increased; in addition, purging water from the GDL under the wide ridges was difficult. Therefore, air was prevented from diffusing into the CL under the wide ridges, thus decreasing the voltage at high current densities. However, the power density curves show that the peak power density of the fuel cell using the locally heated bipolar plate decreased by approximately 0.016 W cm^{-2} at 1.0 A cm^{-2} . This means that normal-temperature performance loss of the locally heated fuel cell is less than 3%. Therefore, the locally heated bipolar plate hardly affects the practical utilization of the PEMFC. Nevertheless, to decrease the voltage loss caused by this bipolar plate, the width of the ridges with the heating wires can be decreased to improve the normal-temperature performance of the fuel cell.

3.2. Effect of local heating on the cold-start performance of PEMFC

This section compares the cold-start performance of the locally heated fuel cell under different heating power densities with the self-starting performance of the fuel cell. Only one heating wire at T1 (Fig. 1(a)) was used to heat the fuel cell under three heating power densities (Table 1) during the cold-start process.

The voltage, impedance, and temperature T3 (Fig. 1(a)) changed for locally heated and self-started fuel cells from -10°C at three heating power densities (P1/P2/P3; Table 1), as shown in Fig. 5(a), and from -20°C at two heating power densities (P'1/P'2; Table 1), as shown in Fig. 5(b). Fig. 5(a) shows that approximately 500 s after cell start-up, the voltage of the fuel cell without local heating decreased to 0 V. When

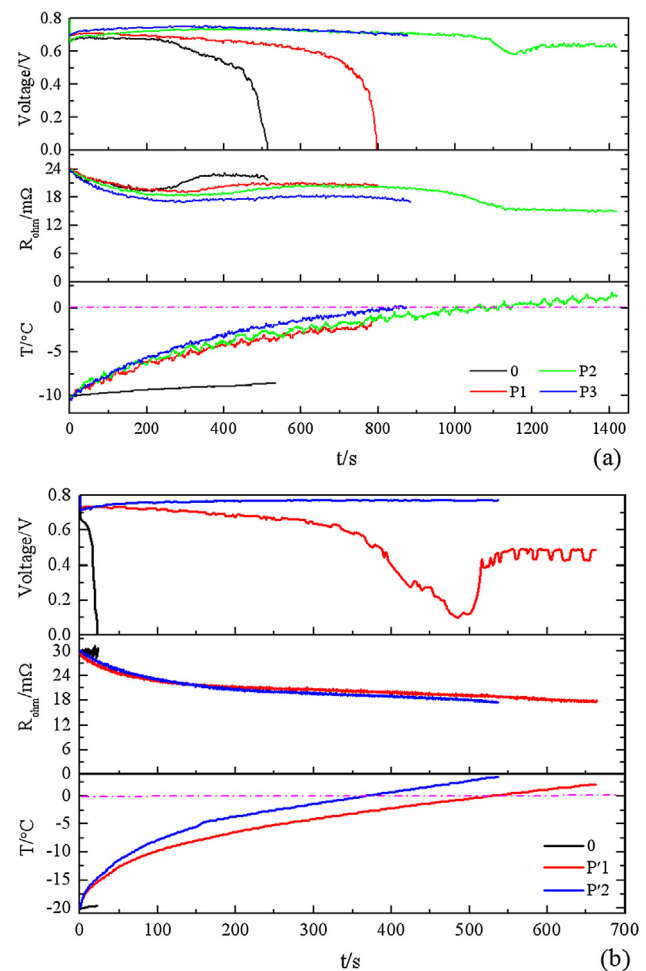


Fig. 5. Changes in voltages, impedances, and temperatures at T3 for the fuel cell initiated at different heating power densities (a) from -10°C and (b) from -20°C .

the heating power density was 47.0 W kg^{-1} , the fuel cell ran continuously for approximately 800 s until the voltage dropped to 0 V. Conversely, when the heating power density reached 52.2 W kg^{-1} , the fuel-cell voltage did not decrease to 0 V before its overall temperature increased above 0°C , indicating a successful cold start. However, the voltage was not stable during the cold-start process for low heating power density. That is, when the heating power density was at 60.6 W kg^{-1} , the total temperature of the fuel cell increased above 0°C , 200 s before that for the heating power density of 52.2 W kg^{-1} . In particular, at 60.6 W kg^{-1} , the PEMFC voltage was relatively stable. The aforementioned experimental results clearly illustrate that the locally heated membrane electrode could improve the fuel cell's cold-start capability. For the self-starting fuel cell, the impedance continued to decrease for approximately 200 s from the start-up, as shown in Fig. 5(a), because the PEM was wetted with water produced in the cathode CL to decrease ion transfer resistance. As the electrochemical reaction continued, water that could not be absorbed by the PEM began to flow into the interface between the CL and MPL in the cathode. Water at the CL/MPL interface in the cathode froze after approximately 200 s, increasing the interface electrical contact resistance. Therefore, the fuel-cell impedance increased at that time. For the locally heated fuel cell at a low heating power density (47.0 W kg^{-1}), the impedance increased after approximately 300 s, illustrating that local heating prolonged the starting time of ice formation. However, the fuel cell eventually failed to cold start owing to the gradually rising temperature. When the heating power density was increased to 52.2 W kg^{-1} after 300 s from

start-up, the impedance first increased before decreasing. This decrease in impedance after approximately 1000 s was due to ice melting when the overall temperature of the PEMFC increased above 0 °C. By combining the voltage and impedance changes, we considered that 52.2 W kg⁻¹ might be the lowest heating power density required to successfully cold start the fuel cell in this experiment. When the heating power density was ultimately increased to 60.6 W kg⁻¹, the PEMFC impedance did not appreciably increase owing to the small amount of ice generated in the fuel cell during the cold-start process. Fig. 5(b) shows that when the fuel cell was initiated from -20 °C, the starting voltage of the locally heated fuel cell increased. This increase was due to the rapid increase in the temperature of the local reaction zone, which increased the activation area of the fuel-cell reaction. When the power density of the heating fuel cell exceeded 365.0 W kg⁻¹, the fuel cell could cold start successfully. In summary, the increase in heating power density ensured not only a fast, successful cold start but also a stable output voltage.

3.3. Effect of number of heating wires on cold-start performance

When the heating power density is the same, the temperature-field distribution in the reaction zone is affected by the number of heating wires in the cathode. This change in temperature-field distribution may in turn affect the cold-start performance of the fuel cell. Therefore, the cold-start capability of the fuel cell was analyzed using one and three heating wires under the same total heating power density.

The changes in the voltages, impedances, and cathode GDL temperatures for the fuel cells using one and three heating wires during the cold-start process from -10 °C are shown in Fig. 6 for a heating power density of 52.2 W kg⁻¹. The fuel cell with one heating wire initiated successfully from -10 °C, while the voltage of the fuel cell using three heating wires dropped to 0 V after 1080 s, as shown in Fig. 6(a). This is because the fuel cell using one heating wire had better cold-start capability than that using three heating wires. Fig. 6(b) shows that in the first 200 s, the fuel cell using one heating wire showed a greater decrease in impedance than that using three heating wires, implying that the former had better wetting ability on the PEM. The temperature at T1 (central position of cathode GDL) for the fuel cell using one heating wire increased above 0 °C after approximately 230 s, and certain non-frozen areas existed in the central reaction zone. As the temperature in the central reaction zone continued to increase, these nonfrozen reaction areas also constantly expanded until the temperature at T3 (position at the edge of the cathode GDL) increased above 0 °C after

approximately 1100 s. However, for the fuel cell using three heating wires (Fig. 6(d)), the temperatures at T1 and T2 increased above 0 °C after approximately 400 and 850 s, respectively. In particular, the temperature at T3 remained under 0 °C until the fuel cell failed to start. This is because every wire was powered with less amount of heat to locally heat the membrane electrode under the wide ridges. Therefore, the cathode GDL temperature near the heating wires increased gradually, resulting in the formation of more ice in the cathode. In contrast, in the fuel cell using one heating wire, the wire was powered at a large heat density under the condition that total heating power was constant. Moreover, the central reaction zone generated relatively less ice, as the temperature near the heating wire rapidly increased above 0 °C. The central zone continued with the electrochemical reaction and released heat power to heat other reaction zones. Finally, the fuel cell using one heating wire successfully cold started from -10 °C. Fig. 6(a) shows that the voltage of the PEMFC using one heating wire dropped considerably after approximately 1100 s, started to increase after approximately 1150 s, and stabilized after approximately 1200 s. However, the voltage after 1200 s showed a loss of 0.06 V relative to that before 1100 s. As shown in Fig. 6(c), the temperature at T3 increased above 0 °C after approximately 1150 s, indicating that the ice in the entire reaction zone melted. We deduce that the ice accumulating in the membrane electrode covered a large area of the reaction zone, thus causing a concentration polarization in voltage. Then, the reaction zone previously covered by ice started to react again after approximately 1150 s owing to the melting ice. Thus, the voltage started to increase at that time. However, the melted water at the CL/MPL interface in the cathode could not be removed quickly enough, so the voltage still showed a concentration polarization of 0.06 V.

Fig. 7 shows the effect of the number of heating wires on the cold-start performance of a PEMFC from -10 °C for a local heating power density of 60.6 W kg⁻¹. As shown in Fig. 7(a), the fuel cells using either one or three heating wires could be successfully cold started at 60.6 W kg⁻¹. The fuel cell using one heating wire showed a stable voltage during the cold-start process. In addition, its impedance remained relatively stable after approximately 200 s (Fig. 7(b)) because less ice formed in the fuel cell. Thermocouples T2 and T3 located away from the central heating wire showed similar temperature (Fig. 7(c)), which increased above 0 °C after approximately 800 s, whereas the temperature at T1 (Fig. 1(a)) increased above 0 °C after approximately 150 s. Therefore, ice barely formed in the central reaction zone, and the PEM near the heating wire showed a greater ion-transfer rate because of its higher temperature. In contrast, for the fuel cell with three heating

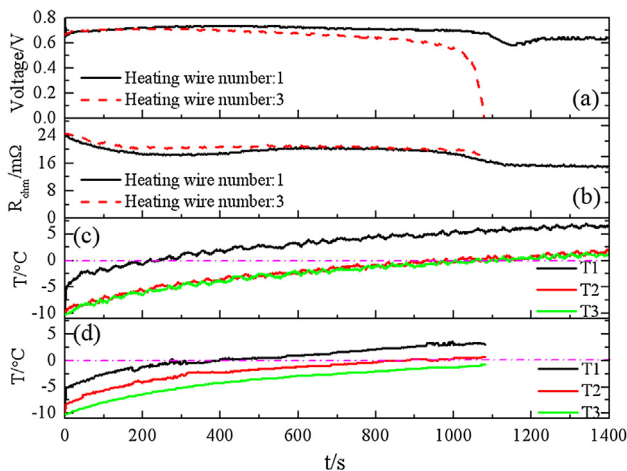


Fig. 6. Effect of the number of heating wires on the cold-start performance (power density: 52.2 W kg⁻¹) with respect to (a) voltage, (b) impedance, (c) cathode GDL temperature (one heating wire), and (d) cathode GDL temperature (three heating wires).

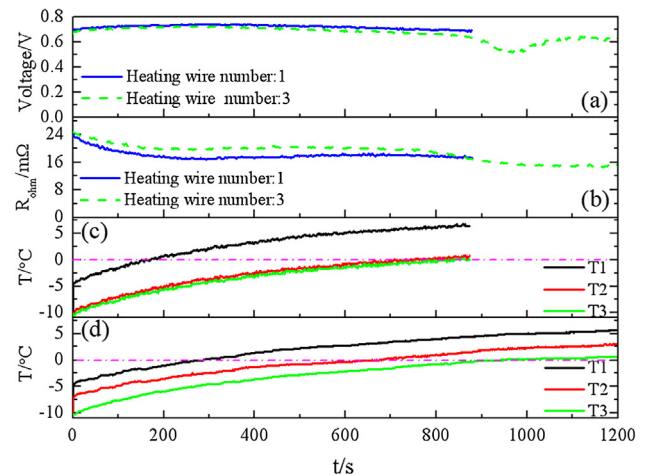


Fig. 7. Effect of the number of heating wires on the cold-start performance (power density: 60.6 W kg⁻¹) with respect to (a) voltage, (b) impedance, (c) cathode GDL temperature (one heating wire), and (d) cathode GDL temperature (three heating wires).

wires, the voltage dropped after approximately 800 s, increased, and then stabilized (as described in the previous section). The impedance for the fuel cell using three heating wires was still larger than that using one heating wire (Fig. 7(b)) when the heating power density was 60.6 W kg^{-1} . The temperatures at T1, T2, and T3 increased above 0°C after approximately 300, 650, and 900 s, respectively, as shown in Fig. 7(d). In particular, the peak temperature at T1 for the fuel cell using three heating wires increased more gradually than that for the fuel cell using one heating wire, resulting in more ice formation and larger electrical contact resistance between the CL and MPL in the cathode central zone. As shown in Fig. 7(b), with ice melting after 800 s, the impedance decreased again for the fuel cell using three heating wires. Then, the melted water decreased the CL/MPL electrical contact resistance in the cathode, resulting in concentration impedance. Moreover, the partially melted water was absorbed by the PEM, which decreased its ion-transfer resistance.

In summary, for a constant heating power density, the fuel cell using one heating wire showed better cold-start performance than that using three heating wires.

4. Conclusions

In this study, a local-heating method was proposed for cold starting a PEMFC by placing heating wires under the partial ridges in the cathode bipolar plate. In addition, the voltage, high-frequency impedance, and cathode GDL temperature for the fuel cell were measured from the initial temperature of -10°C or -20°C to study the effects of different heating power densities and number of heating wires on the cold-start capability of the fuel cell. The main conclusions are as follows.

- (1) Local heating apparently improves the cold-start ability of the fuel cell. The greater the local heating power, the faster the temperature of the reaction zone increases, which is conducive to successfully cold starting a PEMFC. Moreover, compared with other external heating methods (e.g., heating the circulate water or cathode air), directly heating the fuel cell cathode MEA enabled higher energy efficiency during cold starting.
- (2) Insufficient local heating power density causes unstable voltage and even failure of the cold-start process.
- (3) At a constant heating power density, the fuel cell using one heating wire showed better cold-start performance than that using three heating wires. This means that concentrated heating had better effect than dispersed heating, and quickly increasing the local temperature was conducive to a successful cold start.

The local-heating method in this study can be applied to cold start a PEMFC stack for vehicles. Fuel-cell stacks can be rapidly cold started by arranging a certain number of heating wires in the stack to directly heat a partial region of each MEA and setting the power of heating wires according to the ambient temperature.

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